

Hands on practice - Custom Movie Ticket Calculator.

Instead of doing everything at once, we will build this step-by-step. Each task adds exactly one new coding concept so you can see how it works.

The Goal

Build a program that calculates the price of a movie ticket based on the user's age, checks if they have a student discount coupon, and applies a snack package upgrade.

Task	Code
Task 1: Setting up variables and Math (Operators & Output) Concept: Primitive Data Types, Arithmetic Operators, System.out.println We will start with a baseline fixed price and add a snack upgrade using math. 1. Create a file named TicketCalculator.java. 2. Create 2 variable baseprice and comboPrice. 3. Calculate the total price of ticket	<pre> public class TicketCalculator { public static void main(String[] args) { // Step 1: Create variables for pricing double baseTicketPrice = 12.50; double snackComboPrice = 7.25; // Step 2: Use arithmetic operators to find the total double totalCost = baseTicketPrice + snackComboPrice; // Step 3: Print the results to the screen System.out.println("--- Cinema Receipt ---"); System.out.println("Base Ticket: \$" + baseTicketPrice); System.out.println("Snack Combo: \$" + snackComboPrice); System.out.println("Total Cost: \$" + totalCost); } } </pre>
Task 2: Asking User Questions (System Input) Concept: Using Scanner to read keyboard inputs. Right now, the numbers are hardcoded. Let's make it interactive by asking the user for their name and how many tickets they want. 1. Add import java.util.Scanner; to the very top of your file. 2. Take input - customer name, how many tickets. 3. Calculate total price	<pre> import java.util.Scanner; public class TicketCalculator { public static void main(String[] args) { // Create the Scanner tool to read inputs Scanner input = new Scanner(input); System.out.print("Enter your name: "); String customerName = input.nextLine(); // Reads a line of text System.out.print("How many tickets do you want to buy? "); int ticketCount = input.nextInt(); // Reads a whole integer number double singleTicketPrice = 12.50; // Multiply ticket count by the price double totalCost = ticketCount * singleTicketPrice; System.out.println("\n--- Receipt for " + customerName + " ---"); System.out.println("Tickets bought: " + ticketCount); System.out.println("Total Balance due: \$" + totalCost); input.close(); } } </pre>

Task 3: Adding Age Discounts (Control Flow / Decision Making)

Concept: Using if-else and comparison operators (<, >=) to change prices.

Let's make the pricing dynamic. Kids under 12 get a discount, and seniors (65+) get a discount.

Everyone else pays full price.

Update your code.

```
import java.util.Scanner;

public class TicketCalculator {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);
        System.out.print("Enter your age: ");
        int age = input.nextInt();
        double finalTicketPrice;

        // Decision logic using comparison operators
        if (age < 12) {
            System.out.println("Child discount applied!");
            finalTicketPrice = 8.00;
        } else if (age >= 65) {
            System.out.println("Senior discount applied!");
            finalTicketPrice = 9.00;
        } else {
            System.out.println("Regular adult pricing.");
            finalTicketPrice = 13.50;
        }

        System.out.println("Your ticket price is: $" + finalTicketPrice);
        input.close();
    }
}
```

Task 4: Organizing with Functions

Concept: Moving the calculation logic out of main into a reusable function block.

As your code gets bigger, it gets messy. Let's build a dedicated function that calculates the price for us.

```
import java.util.Scanner;

public class TicketCalculator {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);
        System.out.print("Enter your age: ");
        int userAge = input.nextInt();
        // We CALL our custom function here and pass it the user's age
        double price = determinePrice(userAge);
        System.out.println("Final Calculated Price: $" + price);
        input.close();
    }

    // Our new standalone function
    public static double determinePrice(int age) {
        if (age < 12) {
            return 8.00; // Instantly exits the function and passes 8.00 back
        } else if (age >= 65) {
            return 9.00;
        } else {
            return 13.50;
        }
    }
}
```

Task 5: Let It Run Continuously (Loops)

Concept: Using a while loop to let a family calculate multiple different tickets without restarting the program.

Finally, let's wrap everything inside a loop so the app keeps asking for ages until you type a negative number to quit.

```
import java.util.Scanner;

public class TicketCalculator {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);
        boolean keepRunning = true;
        System.out.println("=== Movie Theater Ticket System ===");
        System.out.println("(Type any negative number like -1 to quit)\n");

        while (keepRunning) {
            System.out.print("Enter moviegoer's age: ");
            int age = input.nextInt();
            // Jump or break condition
            if (age < 0) {
                System.out.println("Exiting system. Have a great movie!");
                keepRunning = false; // This breaks the loop condition
            } else {
                double ticketPrice = determinePrice(age);
                System.out.println("Ticket Price: $" + ticketPrice + "\n");
            }
        }
        input.close();
    }

    public static double determinePrice(int age) {
        if (age < 12) return 8.00;
        if (age >= 65) return 9.00;
        return 13.50;
    }
}
```

Challenge

1. Change the pricing function so that teenagers (ages 13 to 19) get a unique ticket price of \$10.00.
2. Add a simple System.out.println message inside the loop to track how many total tickets have been calculated.